Guangya (Wayne) Wan

Ph.D. Candidate in Data Science, University of Virginia

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Research Interests

Self-improving and recursively self-improving AI; automated AI research (auto-research agents); long-horizon LLM agents; multimodal LLM agents, computer-use agents, and visual tool use; System 1 / System 2 reasoning and adaptive test-time compute; LLM post-training (RL); LLM agents for robotics, physical AI, and financial trading; trustworthy AI.

Education

University of Virginia — Charlottesville, VA
Ph.D. in Data Science, Advisor: Prof. Sheng Li — Aug 2023 – December 2027 (expected)
Focus: LLM Reasoning, LLM Agents, Trustworthy AI

Harvard University — Cambridge, MA
M.S. in Biostatistics, GPA: 3.90/4.0 — Aug 2021 – May 2023

University of Illinois at Urbana-Champaign — Urbana, IL
B.S. in Statistics, *Magna Cum Laude*, GPA: 3.94/4.0; Minor: Computer Science & Mathematics — Sep 2017 – Dec 2020

Research & Industry Experience

Waymo – Research Intern — Mountain View, CA | May 2026 – Sep 2026

- Built an **autonomous AI research agent** that improves Waymo's motion-forecasting models end to end without human supervision: it searches related literature, proposes code patches to the model, launches training experiments, analyzes the results, and iterates in a closed loop.
- Integrated the agent with Waymo's internal model-development pipeline, so each research iteration runs real training and evaluation on autonomous-driving data.
- In a separate project, helped design **Group Relative Policy Optimization (GRPO)** training for LLMs that predict driving action labels from scenes; designed the **reward signals** and built the **image-to-text data-cleaning pipeline** that converts camera images into text summaries for language-based training.

Google Cloud – Vertex AI – Student Researcher — Mountain View, CA | May 2025 – Mar 2026

- Developed production-level code for **DeepResearch Agent**.
- Led **SkillForge** (first author, **EMNLP 2026**), a training-free framework that pairs successful and failed runs of the same task to distill validated procedural skills, and loads them on demand at inference time, making long-horizon agents more consistent and efficient across repeated runs.
- Led **COMPASS** (first author, **ACL 2026**), a framework that improves long-horizon agent reasoning by maintaining and evolving the agent's context as a task unfolds.
- Researched test-time scaling and reinforcement learning methods for long-horizon agents.

University of Virginia – Graduate Research Assistant — Charlottesville, VA | Aug 2023 – Present

- Designed **BEACON** (first author, **Findings of EMNLP 2026**), which casts reward-guided LLM sampling as a Bayesian optimal-stopping problem, deciding when to stop generating samples to reduce inference cost.
- Proposed **Reasoning-Aware Self-Consistency** (first author, **NAACL 2025**), which uses the quality of reasoning paths, not just final answers, to weight votes and stop sampling early, making self-consistency more efficient.
- Proposed **Derailer-Rerailer** (first author, **Findings of ACL 2025**), an adaptive verification framework that checks whether a model's reasoning is stable and triggers verification and correction only when needed.
- Co-developed **WiseMind** (co-first author, ***npj Digital Medicine*** 2026), a knowledge-guided multi-agent framework for accurate and empathetic psychiatric diagnosis.

Selected Earlier Experience — 2019 – 2023

Data Scientist Intern, **Chewy**; ML Research Intern, **Boston Children's Hospital (CHIP)**; ML Research Intern, **NCSA SPIN**. Work included forecasting, fairness-aware clinical ML, and HPC-based disease modeling.

Selected Publications

1. **G. Wan**, H. Wang, X. Ling, M. Ling, X. Ren, S. Li. "SkillForge: Learning from Success–Failure Boundaries for Consistent Long-Horizon Agents." **EMNLP 2026** (to appear).
2. **G. Wan**, M. Ling, X. Ren, R. Han, S. Li, Z. Zhang. "COMPASS: Enhancing Agent Long-Horizon Reasoning with Evolving Context." **ACL 2026**. **Paper Link**
3. **G. Wan**, Z.S. Xu, S. Zorc, M. Baucells, M. Hu, H. Wang, S. Li. "BEACON: Bayesian Optimal Stopping for Efficient LLM Sampling." **Findings of EMNLP 2026** (to appear). **Paper Link**
4. **G. Wan**, Y. Wu, J. Chen, S. Li. "Reasoning-Aware Self-Consistency: Leveraging Reasoning Paths for Efficient LLM Sampling." **NAACL 2025**. **Paper Link**
5. **G. Wan**, Y. Wu, H. Wang, S. Zhao, J. Chen, S. Li. "Derailer-Rerailer: Adaptive Verification for Efficient and Reliable Language Model Reasoning." **Findings of ACL 2025**. **Paper Link**
6. **G. Wan**, Y. Lu, Y. Wu, M. Hu, S. Li. "Large Language Models for Causal Discovery: Current Landscape and Future Directions." **IJCAI 2025** Survey Track. **Paper Link**
7. Y. Wu*, **G. Wan***, J. Li, S. Zhao, L. Ma, T. Ye, I. Pop, Y. Zhang, J. Chen. "WiseMind: A Knowledge-Guided Multi-Agent Framework for Accurate and Empathetic Psychiatric Diagnosis." *npj Digital Medicine*, 2026. **Paper Link**
8. R. Zhou*, **G. Wan***, S. Gabriel, S. Li, A. Gates, M. Sap, T. Hartvigsen. "Disparities in LLM Reasoning Accuracy and Explanations: A Case Study on African American English." **COLM 2025** MELT Workshop. **Paper Link**
9. X. Li, M. Gao, Y. Hao, T. Li, **G. Wan**, Z. Wang, Y. Wang, X. Chen. "MedGUIDE: Benchmarking Clinical Decision-Making in Large Language Models." **NeurIPS 2025** GenAI4Health Workshop. **Paper Link**
10. **G. Wan**, J. Allen, W. Ge, et al. "Two-step LightGBM for Human West Nile Virus Risk in Chicago." *PLOS ONE* 19(1): e0296283, 2024. **Paper Link**
11. J.A. McCoy, L.D. Levine, **G. Wan**, et al. "Deep Learning on Intrapartum FHR to Predict Acidemia at Birth." *American Journal of Obstetrics and Gynecology*, 2024. **Paper Link**
12. W.G. La Cava, E. Lett, **G. Wan**. "Fair Admission Risk Prediction with Proportional Multicalibration." **CHIL 2023**. **Paper Link**

* Equal contribution.

Honors & Awards

Computational Health Informatics Fellowship, Harvard University (2022); Magna Cum Laude, UIUC (2020); SPIN Fellowship, NCSA (2020); Best Presentation Nominee, Canadian Undergraduate Mathematics Conference (2020); Edmund J. James Scholar, UIUC (2018–2020).

Academic Service

Conference Reviewer (20+ reviews): NeurIPS (2024, 2025), ICLR (2025, 2026), ICML 2025, ACL 2026, COLM 2025, AISTATS 2025
Journal Reviewer: IEEE Computational Intelligence Magazine
Teaching Assistant: UVA (CS 5012, DS 6310, DS 6030); Harvard (CS 109B)

Technical Skills

AI-Native Development: Claude Code, OpenAI Codex, Gemini CLI, Cursor, GitHub Copilot, MCP
LLM Training & Post-Training: PyTorch, JAX, HF Transformers, TRL, PEFT/LoRA, DeepSpeed, FSDP, VeRL
LLM Inference & Serving: vLLM, SGLang, Hugging Face TGI, OpenAI / Anthropic / Gemini SDKs
Agents & Evaluation: Google ADK, LangGraph, DSPy, lm-eval-harness, τ^2-Bench, Terminal-Bench
Cloud & Systems: Google Cloud (Vertex AI), AWS, Docker, Kubernetes, Slurm, Git, Unix, Weights & Biases
Data: Python, SQL, PySpark, Pandas, Snowflake